

Homework 6

1. Using Taylor series, compute the following limits:

(a) $\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^5}.$

(b) $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x - 3x^2/2}{x^4}.$

2. Prove that the Taylor series of $\ln(1+x)$ around 0 is

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}.$$

3. Using Taylor series, prove that the forward difference $(f(x+h) - f(x))/h$ approximates the derivative $f'(x)$ with an error of order h (first order approximation), i.e.

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h),$$

and that the centered difference $(f(x+h) - f(x-h))/2h$ approximates the derivative $f'(x)$ with an error of order h^2 (second order approximation), i.e.

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2).$$

Here the Big Oh notation $e_1(h) = O(e_2(h))$ means that $\lim_{h \rightarrow 0} \frac{e_1(h)}{e_2(h)} \in (0, \infty)$.

Python:

Choose a particular function f , a point x and compute exactly $f'(x)$. By taking a range of small values h , show that the errors when approximating $f'(x)$ with the finite differences above are proportional to h and h^2 , respectively.

Explain why after h becomes small enough the errors start to increase.